

Lecture Notes

#date/2025/09/26

Charge is a physical property of an object like math

Protons → positively charged

Electrons → negatively charged

atom → # protons = # electrons

atoms naturally electrically uncharged, unless they are charged

because nucleus can't be broken, an atom can only be charged into an ion by adding or removing electrons

unit for charge on an object: q or Q

SI unit for charge: Coulomb: C

point charge, has no dimension (weight/thickness)

an electric field is the effect of other objects from a charge

point charge is a source of an electric field, and each field is unique to each object's property

it's very important to only couple interactions between fields and their related properties, not properties that aren't related to that field. for example you can't interact mass and electrical fields, or charge and gravitational fields, you must interact charge with electrical fields

when an object has a charge it will have an electric field, which gets stronger with the distance from and strength of a charge

an e-field is due to a positive or negative charge

when drawing a field the arrow should never start at the charge, it should point in the direction of the field, for example the arrow points away from a positively charged point and towards a negatively charged point

The magnitude of the E-field is due to point charges or spherically symmetric charges but outside of the charge

$$|\vec{E}| = \frac{k|q|}{r^2}(\pm\hat{r})$$

k is a constant of electrostatic

$$k = 9 \times 10^9 \frac{Nm^2}{C} \text{ (in a vacuum or air)}$$

$$k = \frac{1}{4\pi\epsilon_0} \text{ (in a vacuum)}$$

ϵ_0 : permittivity of air/vacuum

r : distance from point charge or center of the spherical symmetric charge

$\hat{r}$: unit vector along r

$+\hat{r}$: pointing away from source charge

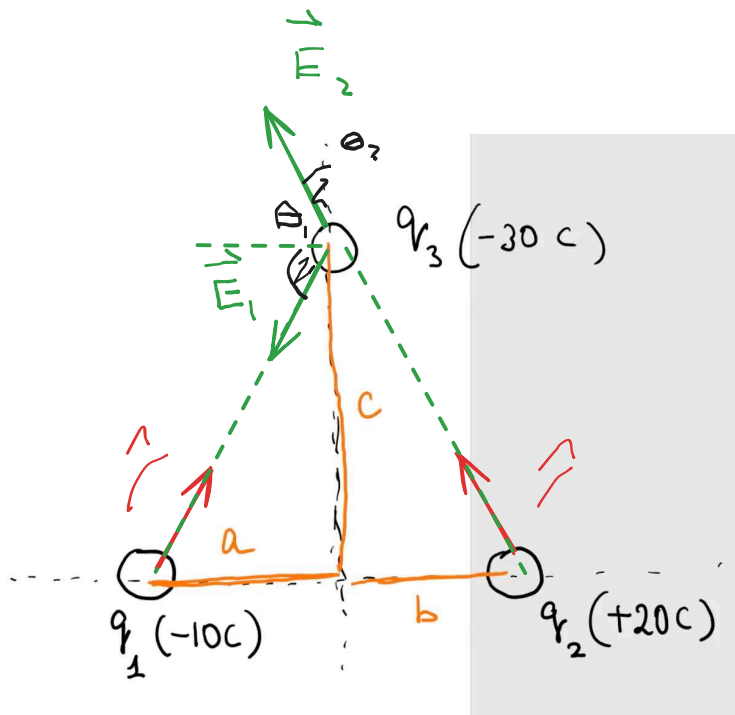
$-\hat{r}$ pointing toward source charge

EX

$a = 2m$; $b = 3m$; $c = 4m$

calculate the E-field on q_3

the E-field on q_3 we only need E_1, E_2



Target: $|\vec{E}|, \theta_E$

Method

$$|\vec{E}| = \sqrt{E_x^2 + E_y^2}$$

$$\begin{aligned}
E_x &= \vec{E}_{1x} + \vec{E}_{2x} \\
&= -E_1 \cos \theta_1 - E_2 \sin \theta_2 \\
&= -E_1 \frac{a}{r_1} - E_2 \frac{b}{r_2} \\
\vec{E}_x &= -E_1 \frac{a}{r_1} - E_2 \frac{b}{r_2} \\
&= -\frac{k|q_1|}{r_1^2} \left(\frac{a}{r_1} \right) - \frac{k|q_2|}{r_2^2} \left(\frac{b}{r_2} \right) \\
&= -9 \times 10^9 \left(\underbrace{\frac{10 \cdot 2}{89.44}}_{r_1^3} + \underbrace{\frac{20 \cdot 3}{125}}_{r_2^3} \right) \\
\vec{E}_x &= -6.33 \times 10^9 \frac{N}{C}
\end{aligned}$$

$$\begin{aligned}
\vec{E}_y &= \vec{E}_{1y} + \vec{E}_{2y} \\
&= -E_1 \sin \theta_1 + E_2 \cos \theta_2 \\
&= -E_1 \left(\frac{c}{r_1} \right) + E_2 \left(\frac{c}{r_2} \right) \\
&= -\frac{k|q_1|}{r_1^2} \left(\frac{c}{r_1} \right) + \frac{k|q_2|}{r_2^2} \left(\frac{c}{r_2} \right) \\
&= 9 \times 10^9 \left(-\frac{(+10C) \times 4}{89.44} + \frac{20 \times 4}{125} \right) \\
\vec{E}_y &= 1.73 \times 10^9 \frac{N}{C}
\end{aligned}$$

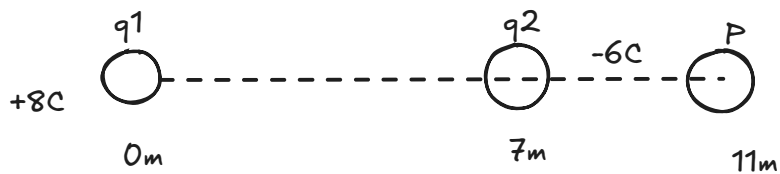
$$\begin{aligned}
\tan \theta_E &= \frac{|E_x|}{|E_y|} \\
\theta_E &= \tan^{-1} \left(\frac{|E_x|}{|E_y|} \right) \\
&= \tan^{-1} \left(\frac{6.33}{1.73} \right) \\
&= 74.71 \\
&\text{Counter Clockwise from y-Axis}
\end{aligned}$$

$$\begin{aligned}
\vec{F}_{\text{on } 3} &= q_3 E_3 \\
|F_3| &= |q_3 E_3| \\
&= 30C \times 6.56 \times 10^9 \frac{N}{C} \\
&= 196.8 \times 10^9 N \\
&= 1.968 \times 10^{11} N \\
&74.71 \text{ deg counter cw from y-axis}
\end{aligned}$$

#date/2025/09/29

warm up

Calculate the electric force at point C



$$\begin{aligned}
 |\vec{E}_p| &= E_1 + E_2 \\
 E_p &= \frac{k|q_1|}{r^2}(\pm\hat{r}) + \frac{k|q_2|}{r^2}(\pm\hat{r}) \\
 &= \frac{k|+8|}{11^2}(\pm\hat{r}) + \frac{k|-6|}{7^2}(\pm\hat{r}) \\
 &= k\left(\frac{8}{11^2} + \frac{6}{7^2}\right) \\
 E_p &= 1.697 \times 10^9 \\
 F_p &= q_p E_p
 \end{aligned}$$

q_p is unknown

Ions

what is the charge on doubly ionized lithium

Lithium Atomic Number: 3

double ionization: ± 2 electrons

+2e charge when electrons are removed (less negative)

-2e charge when electrons are added (more negative)

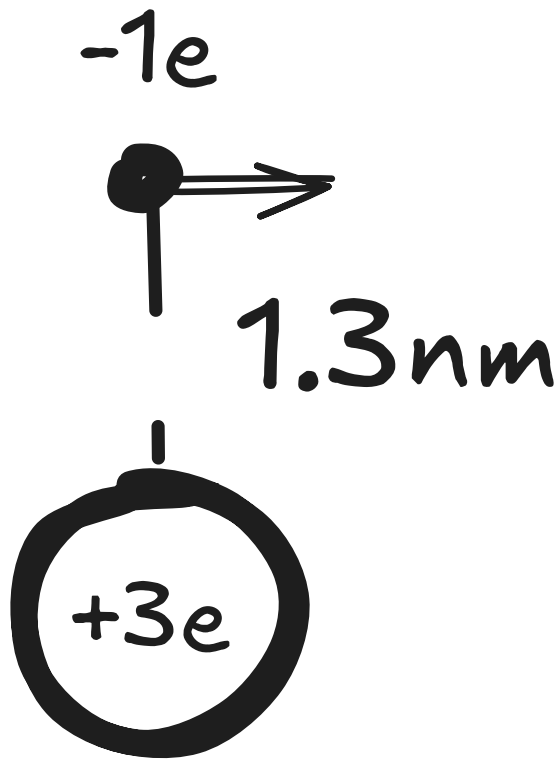
the charge of 1 electron is constant

$$1e = 1.602 \times 10^{-19}$$

$$q_L = \pm 2(1.602 \times 10^{-19})$$

Exercise

Q: Calculate the number of rotations the electron in doubly ionized lithium is making per second around the nucleus, assuming it is orbiting at a distance of $1.3nm$ from the nucleus.



Given: $q_L = 3.204 \times 10^{-19}$; $d = 1.3\text{nm} = 1.3 \times 10^{-9}\text{m}$; $m_e = 9.11 \times 10^{-31}$

steps: charge $\rightarrow$ field $\rightarrow$ force $\rightarrow$ accel $\rightarrow$ speed

$$\vec{a}_{rad} = \frac{F_{net(rad)}}{m}$$

$$\begin{aligned} |\vec{E}_L| &= \frac{k|q|}{r^2} (\pm \hat{r}) \\ &= (9 \times 10^9) \frac{3 \times 1.602 \times 10^{-19}}{1.33 \times 10^{-9}} \\ E_L &= 3.25218 \end{aligned}$$

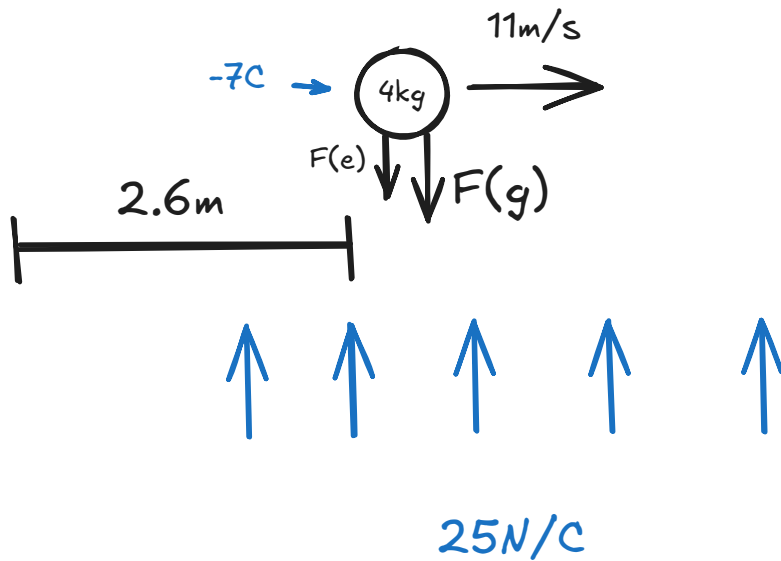
$$\begin{aligned} F &= q_e(E_L) \\ &= 1.602 \times 10^{-19} (3.25218) \\ &= 5.20999 \times 10^{-19} \end{aligned}$$

$$\begin{aligned} \vec{a}_{rad} &= \frac{F_{net(rad)}}{m} \\ \frac{v^2}{r} &= \frac{5.20999 \times 10^{-19}}{9.11 \times 10^{-31}} \end{aligned}$$

#date/2025/10/1

Warm up Problem:

you throw a 4kg ball with a -7C charge, horizontally in air (on earth) at $11\frac{\text{m}}{\text{s}}$ into a region of uniform E-field of $25\frac{\text{N}}{\text{C}}$ pointing upward. In what direction is the ball moving (w.r.t. horizontal) after traveling 2.6m horizontally



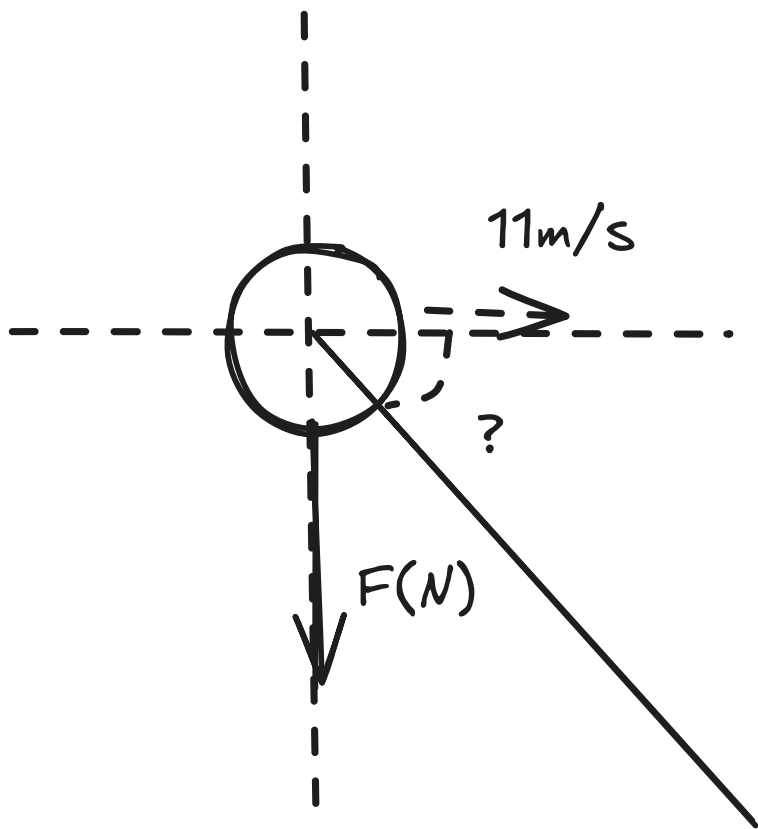
Given: mass: 4kg , charge of object: -7C , V : 11m/s , E-field: 25n/c , distance moved: 2.6m

Target: direction the ball is traveling

The net force will be in the downward direction, because both the force of gravity and the electric force are directed downward

#note a positive charge experiences electrical force in the direction of the field, a negative charged object will experience it in the opposite direction

to get the direction we need to calculate the total horizontal and vertical displacement



$$\vec{V}_{f_x} = \vec{V}_{0_x} + \cancel{\vec{a}_x} t$$

$$\vec{V}_{f_x} = \vec{V}_{0_x}$$

$$\vec{V}_{f_y} = \vec{V}_{t_{0_y}} + \vec{a}_y t$$

$$\begin{aligned} \vec{a}_y &= \frac{\vec{F}_{net(y)}}{m} \\ &= \frac{\vec{F}_{E(y)} + \vec{F}_{g(y)}}{m} \\ &= \frac{-|qE| - |mg|}{m} \\ &= -\frac{(7 \times 25 + 4 \times 9.81)}{4} \\ \vec{a}_y &= -53.56 \frac{m}{s^2} \end{aligned}$$

$$\Delta \vec{x} = \vec{V}_{0_x} t + \cancel{\frac{1}{2} \vec{a}_x t^2}$$

$$D = v_0 t$$

$$t = \frac{D}{v_0}$$

$$t = 0.2364s$$

$$\begin{aligned}\vec{V}_{fy} &= \cancel{\vec{V}_{0y}} + \vec{a}_y t \\ &= \left(-53.56 \frac{m}{s^2}\right)(0.2364s) \\ \vec{V}_{fy} &= -12.6616\end{aligned}$$

$$\begin{aligned}\tan(\theta) &= \frac{|\vec{V}_{fy}|}{|\vec{V}_{fx}|} \\ \theta &= \tan^{-1}\left(\frac{|-12.6616 \frac{m}{s}|}{|11 \frac{m}{s}|}\right) \\ \theta &= 0.8555 \text{ rad} \\ \theta &= 49.017^\circ\end{aligned}$$

How to find the E-field due to a charged object

#date/2025/10/03

How to Calculate E-fields due to various types of sources.

it is NOT $\frac{kq}{r^2}$

One method is Gauss' law

$\frac{kq}{r^2}$ is derived from Gauss' law for a point charge

Gauss' Law:

If we fully enclose a source of a charge(Q) or mass(m) with some (any) enclosing surface, then the total Flux of an E-Field due to Q (or g-field due to m) is proportional to the charge enclosed by that surface (or mass)

What is Flux of some stuff?

Flux of any stuff is the total amount of that stuff over some surface.
all forms of Flux are scalars!!

How to calculate Flux of any vector quantity $\vec{V}$

ϕ - Symbol for Flux

$$\phi_{\vec{V}} = \int \vec{V}_{//d\vec{a}} \times d\vec{a}$$

where $\vec{V}_{//d\vec{a}}$ means the component of the vector $\vec{V}$ at a point parallel to a portion of the area ($d\vec{a}$) at that point

because of this we can re-write as a dot product

$$\phi_{\vec{V}} = \int \vec{V} \cdot d\vec{a}$$

$$\phi_{\vec{E}} = \int \vec{E} \cdot d\vec{a}$$

if $\vec{E}_{//d\vec{a}}$ (or $\vec{V}_{//d\vec{a}}$) is uniform over a surface we can drop the integral

then $\phi_{\vec{E}} = \vec{E}_{//d\vec{a}} * \vec{A}$

Exercise: Calculate the total E-flux on this disk

given:

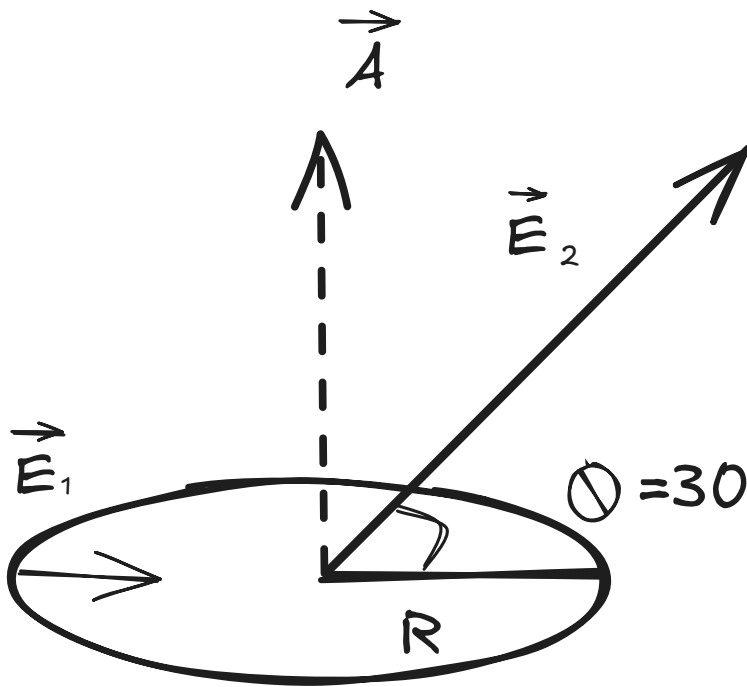
$$R = 3m$$

$$|\vec{E}_1| = 7 \frac{N}{C}$$

$$|\vec{E}_2| = 10 \frac{N}{C}$$

$$\theta_2 = 30^\circ$$

$$A = \pi R^2 = 9\pi$$



$$\begin{aligned} \phi_{\vec{E}} &= \phi_{E_1} + \phi_{E_2} \\ &= \int \vec{E}_1 \cdot d\vec{a} + \int \vec{E}_2 \cdot d\vec{a} \\ &= \int \vec{E}_1 \cdot \hat{n} \times A + \int \vec{E}_2 \cdot \hat{n} \times A \\ &= \int (7) \cdot \cancel{(9\pi) \cos(90)}^0 + \int (10) \cdot (9\pi) \cos(30) \\ \phi_{\vec{E}} &= \phi_{\vec{E}_2} = 90\pi \left(\frac{\sqrt{3}}{2} \right) \\ \phi_{\vec{E}} &= 45\pi\sqrt{3} \end{aligned}$$

we know know the quantitative form of Gauss' law:

$$\oint \vec{E}_{\text{enclosed by the surface}} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{\text{enclosed by the surface}}$$

and it turns out that the constant of proportionality are $4\pi k$ and $4\pi G$

ie

$$\oint \vec{E}_{(\text{enc})} = 4\pi k Q_{\text{enc}}$$

where $k = 9 \times 10^9$ in a vacuum or air, and corresponds to the electrical permittivity $[\epsilon]$ of the medium through $k = \frac{1}{4\pi\epsilon}$

How do we derive $\vec{E}$ due to a point charge

at a point p a distance r away from the source

ANY surface you pick will satisfy gauss' law as long as it encloses the source, so we can choose a surface on which the calculation will be easiest

#date/2025/10/06

how to use Gauss's law to find E-fields due to various sources

1. Identify points of interest and imagine passing a Gaussian surface (imaginary surface, your choice) through the point of interest fully enclosing the source
2. choose a shape for the surface so that $\vec{E} \cdot d\vec{a}$ is constant over that surface so the LHS integral becomes very easy
3. ensure to only use the amount of charge inside the gaussian surface
4. then solve Gauss's law relationship to find the E-field

5.

$$\oint \vec{E} \cdot d\vec{a} = 4\pi k Q_{\text{enc}}$$

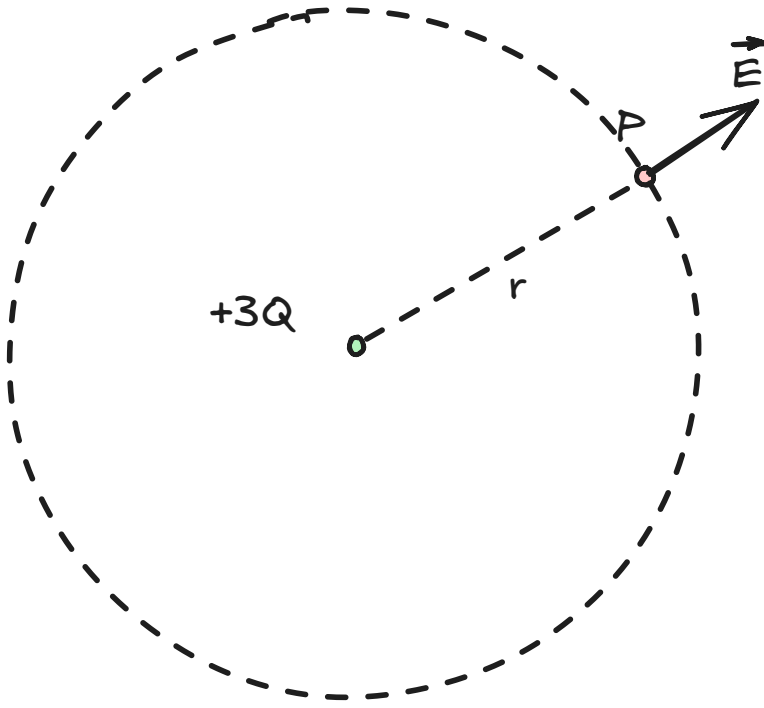
source 1: point charge (or outside spherically symmetric source)

Q: find E-field at P a distance (r) away from +Q:

Target: $E(r)$

Method: $\oint \vec{E} \cdot d\vec{a} = 4\pi k Q_{\text{enc}}$

Answer: $12\pi k Q_{\text{enc}}$



$$E_{\parallel da} A_{\text{sphere}} = 4\pi k(+Q)$$

$$E(4\pi r^2) = 4\pi k(Q)$$

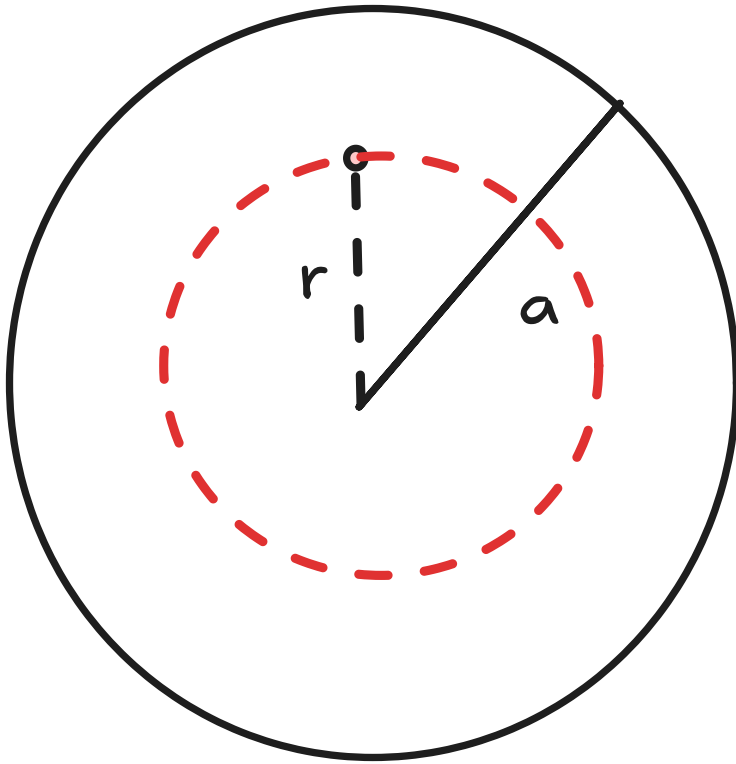
$$E = \frac{k(+Q)}{r^2}$$

Source 2: Inside a non-conducting ball of charge where the charge is distributed non-uniformly with a volume density $\rho(r)$ given as:

$$\begin{cases} \rho(r) = C\left(\frac{r}{a}\right)^5 & \text{for } r \leq a \\ \rho(r) = 0 & \text{for } r > a \end{cases}$$

where a is the radius of the sphere and C is some constant

Q: find $E(r)$ for $r < a$



first we choose our point r and a gaussian surface enclosing that point

$$\oint \vec{E}_{\parallel d\vec{a}} * d\vec{a} = 4\pi k Q_{enc}$$

$$E * A_{sphere} = 4\pi k Q_{enc}$$

$$E(4\pi r^2) =$$

to find Q_{enc} from $\rho(r)$:

$\rho(r)$ is volume charge density

$$\Rightarrow \rho(r) = \frac{dq}{dV}$$

this implies

$$\Rightarrow dq = \rho(r)dV$$

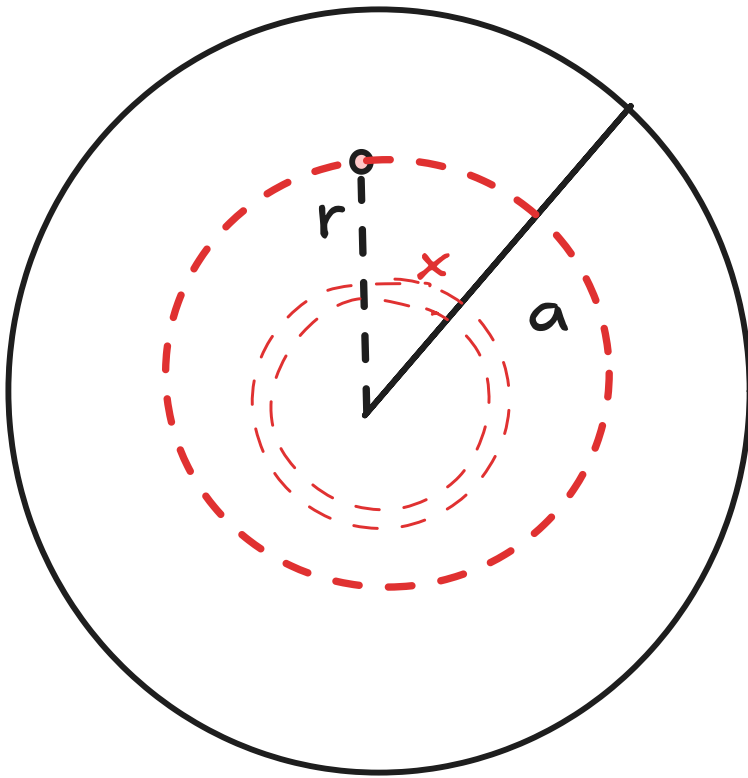
$$\Rightarrow \int dq = \int \rho(r)dV$$

$$\Rightarrow Q_{enc} = \int_0^r \rho(x)dV$$

we need to select an x thats somewhere in $0 \leq x \leq r$

and we can use x to make our Volume differential by drawing a very thin shell of radius x (we pick a shell because the charge changes radially, so if it were too thick the charge would not be uniform)

like:



we can express the volume as $4\pi x^2 dx$
giving

$$\begin{aligned}
 Q_{enc} &= \int_0^r \rho(x) 4\pi x^2 dx \\
 &= \int_0^r C \left(\frac{x}{a} \right)^5 4\pi x^2 dx \\
 &= \frac{4\pi C}{a^5} \int_0^r x^7 dx \\
 &= \frac{4\pi C}{a^5} \left[\frac{x^8}{8} \right]_0^r \\
 &= \frac{4\pi C}{8a^5} r^8
 \end{aligned}$$

now we can substitute this back in to the equation for the E-field

$$\begin{aligned}
 \oint \vec{E}_{\parallel d\vec{a}} * d\vec{a} &= 4\pi k Q_{enc} \\
 E * A_{\text{sphere}} &= 4\pi k Q_{enc} \\
 E(4\pi r^2) &= 4\pi k \left(\frac{4\pi C}{8a^5} r^8 \right) \\
 E(r < a) &= \frac{4\pi k C}{8a^5} r^6
 \end{aligned}$$

to find $E(r > a)$ we are looking for the charge outside the ball I.E. the charge of the whole ball, we cannot integrate outside of a because the density outside of the ball = 0

instead we integrate Q_{enc} for $0 \rightarrow a$ giving us the total charge inside the ball

#date/2025/10/08

Cont Prev Lecture: Calculating E-Field outside ($r > a$)

you have already shown for $r \leq a$ that:

$$\oint \vec{E}_{\parallel d\vec{a}} * d\vec{a} = 4\pi k Q_{enc}$$

$$E(4\pi r^2) = 4\pi k Q_{enc}$$

and that

$$Q_{enc} = \int_0^r \rho(r) 4\pi r^2 dr$$

$$Q_{enc} = \underbrace{\frac{4\pi C}{a^5} \left[\frac{r^8}{8} \right]_0^r}_{\beta}$$

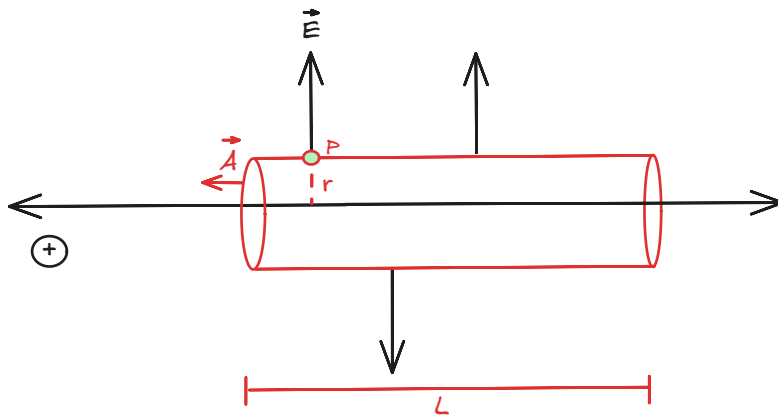
then write for $r > a$ we use $Q_{enc} = \int_0^a \rho(r) 4\pi r^2 dr$

we show which form β becomes

$$E(r > a) = \frac{k}{r^2} \left[\frac{4\pi C}{8a^5} (a^8 - 0^8) \right]$$

$$E(r > a) = \frac{4\pi ck}{8r^2} a^3$$

Source 3: The E-Field due to an infinitely long line of charge



the challenge with an infinite charge like this is finding a gaussian surface that can properly enclose this

we choose a cylinder of to encompass the charge. because if we were to use an infinitely long cylinder we instead choose one of length L

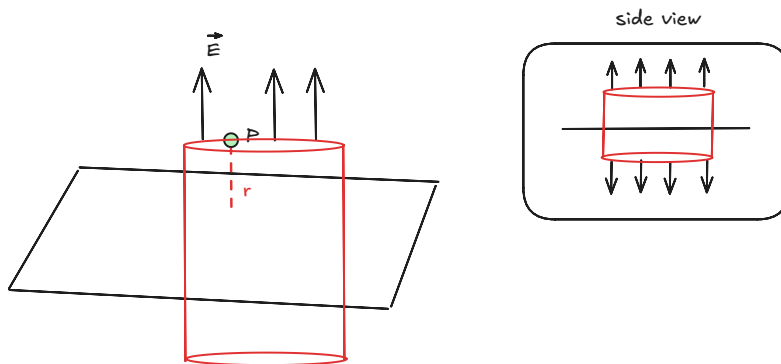
the flux from the cap will be 0 because the area vector is perpendicular to the direction of the charge

for an infinite line of charge we provide charge information using charge density. since the charge is on a line we will define the Linear Charge Density λ : $\lambda = \frac{dq}{dl} = \frac{Q}{L} \implies Q = \lambda L$

$$\begin{aligned}\phi_{enc} &= 4\pi k Q_{enc} \\ \cancel{\phi_{cap_1}}^0 + \cancel{\phi_{cap_2}}^0 + \phi_{cyl} &= 4\pi k Q_{enc} \\ \oint E_{||} da &= 4\pi k Q_{enc} \\ EA_{cyl} &= 4\pi k Q_{enc} = 4\pi k Q_{enc} \\ E[2\pi r L] &= 4\pi k [\lambda L] \\ E(r) &= \frac{2k\lambda}{r}\end{aligned}$$

Source 4: Infinite (or very large) non-conducting sheet of charge

find E(r) at p



$$\begin{aligned}\phi_{enc} &= 4\pi k Q_{enc} \\ \phi_{cap_1} + \phi_{cap_2} + \cancel{\phi_{cyl}}^0 &= 4\pi k Q_{enc} \\ E(A) + E(A) &= 4\pi k Q_{enc} \\ 2EA_{cap} &= 4\pi k Q_{enc} \\ E &= \frac{4\pi k Q_{enc}}{2A_{cap}} \\ E &= \frac{2\pi k Q_{enc}}{A}\end{aligned}$$

#date/2025/10/10

Conductors and Non conductors

Conductors:

Materials made of atoms where outermost valence electrons are loosely-bound to the nucleus. these electrons can get very easily detached from an atom (due to stimulus such as an external E-field, Thermal excitation, etc.) and become available as a (sea of shared electrons) for the material

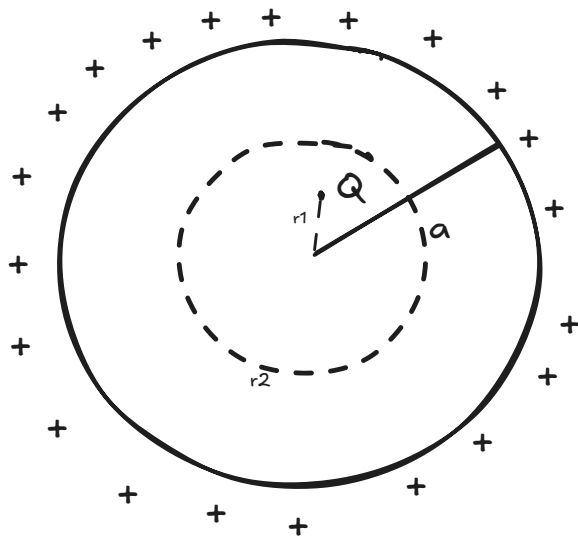
Consequently, these electrons AND any externally added charges positive or negative can move freely within the material in response to an external stimulus

Non-Conductors:

valence electrons are tightly bound to the atoms

Consider a ball of IRON

excess charges
migrate to the
surface of a
conductor



suppose you come in and dump $+23\text{pC}$ of charge at $r_1 = \frac{a}{4}$ what will be the E-field at $r_2 = \frac{a}{2}$
given:

$$Q = +23\text{pC} = +23 \times 10^{-12}$$

$$r_1 = \frac{a}{4}$$

$$r_2 = \frac{a}{2}$$

Target: $E(r_2)$

method:

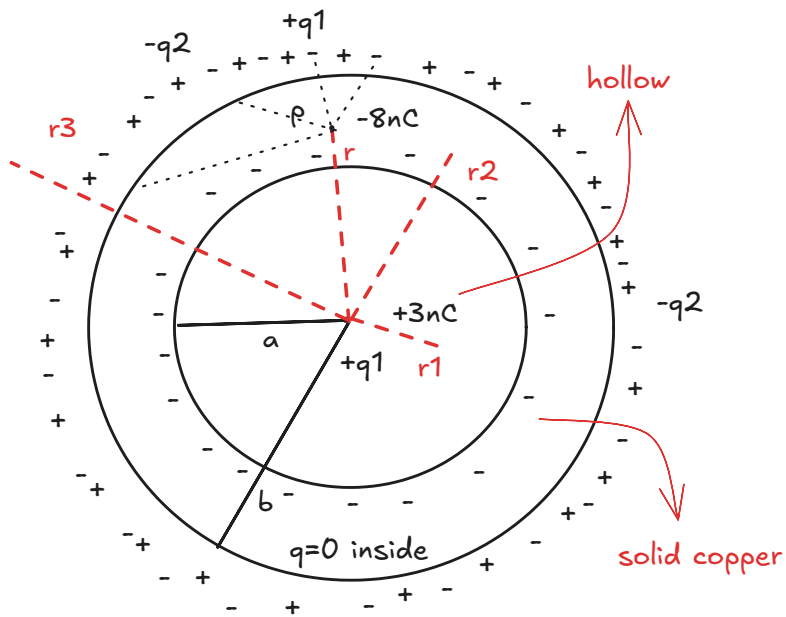
$$\oint \vec{E}_{\parallel d\vec{a}} = 4\pi k Q_{\text{enc}}$$

$$E(4\pi r_2^2) = 4\pi k \cancel{Q_{\text{enc}}}^0$$

$$\implies E = 0 \text{ inside ANY conductor}$$

Ex

you place a $+3nC$ point charge at the center of a hollow copper shell you then inject $-8nC$ of a charge at point P within the shell



given

$q_1 = +3nC$; $q_2 = -8nC$ at $r = 5mm$; $a = 3mm$; $b = 9mm$

calculate:

E_1 at $r_1 = 2mm$; E_2 at $r_2 = 6mm$; E_3 at $r_3 = 12mm$

1. $E(r_1)$:

$$\oint E_{||da} = 4\pi k Q_{enc}$$

$$E(4\pi r_1^2) = 4\pi k q_1$$

$$\Rightarrow E = \frac{kq}{r_1^2}$$

$$r_1 = 4 \times 10^{-6}; q_1 = +3 \times 10^{-9}C$$

2. $E(r_2)$

$$E(4\pi r_2^2) = 4\pi k \cancel{q_{enc}}^0$$

$$E(r_2) = 0$$

because r_2 is inside a conductor

3. $E(r_3)$

$$E(4\pi r_3^2) = 4\pi k Q_{enc}$$

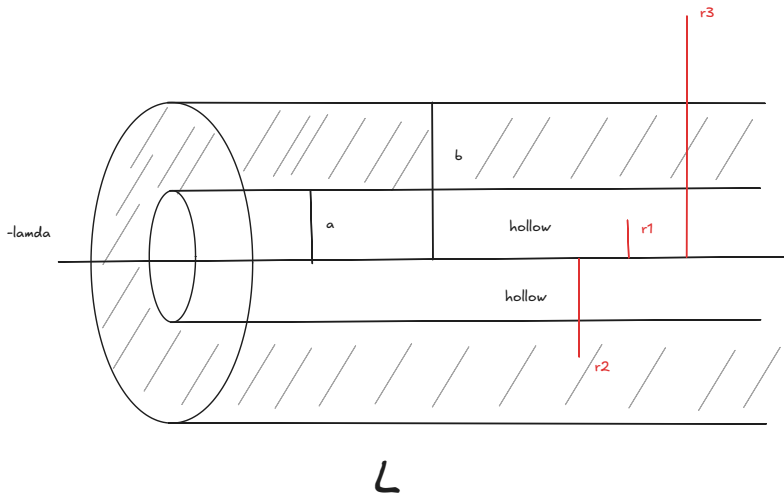
$$E(r_2) = \frac{k q_{enc}}{r_3^2}$$

$$q_{enc} = q_1 + (-q_2) \\ = -5nC$$

Example:

Given $-\lambda$ in wire, a , b , L

now you come in and dump $+Q_{ex}$ of excess charge on the copper shell.



Calculate:

$$E_1(r < a)$$

$$E_2(a \leq r \leq b)$$

$$E_3(r > b)$$

$$E_1 A_1 = 4\pi k Q_{enc} \\ E_1 (2\pi r_1 L) = 4\pi k Q_{enc} \\ \Rightarrow E(r_1) = \frac{2k Q_{enc}}{r_1 L} \\ = \frac{2k \lambda_{enc}}{r_1}$$

$$E_2(a \leq r \leq b) = 0$$

since r is inside the conductor

$$E(r_2) = \frac{2k}{r_3} \lambda_{enc}$$

$$\lambda_{enc} = -\lambda_{wire} + \lambda_{due\ to\ Q_{enc}}$$

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recap:

$$E_{in} = \frac{k\pi C}{2a^6} r^7$$

$$E_{out} = \frac{k\pi C a^2}{2} \frac{1}{r}$$

Method:

$$V(r_1) - V(r_2) = - \left[\int_{r_2}^a E_{out} dr + \int_a^{r_1} E_{in} dr \right] \left\{ \begin{array}{l} E_{in} = D_1 r^7 \\ E_{out} = \frac{D_2}{r} \\ D_1 = \frac{k\pi C}{2a^6} \\ D_2 = \frac{k\pi C a^2}{2} \end{array} \right.$$

$$= - \left[D_2 \int_{r_2}^a \frac{dr}{r} + D_1 \int_a^{r_1} r^7 dr \right]$$

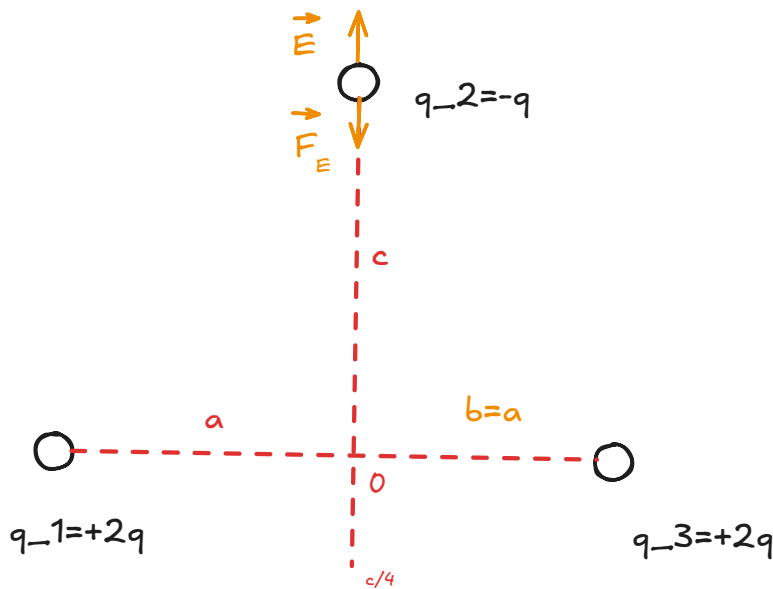
$$= - \left[D_2 \ln \left(\frac{a}{r} \right) + \frac{D_1}{8} [r^8 - a^8] \right]$$

$$\underbrace{V(r)}_{r < a} - V(3a) = - \left[\frac{k\pi C a^2}{2} \ln \left(\frac{1}{3} \right) + \frac{k\pi C}{16a^6} [r^8 - a^8] \right]$$

$$= - \frac{k\pi C a^2}{2} \left[\ln \left(\frac{1}{3} \right) + \frac{1}{8a^8} [r^8 - a^8] \right]$$

Example:

3 pt charges are placed as shown. if you hold q_1 and q_2 fixed and release q_2 from rest



1. what will q_2 do

because of the symmetry, q_2 will move straight down, and then oscillate back and forth across the line

2. if $m_1 = m$; $m_2 = 3m$ and $m_3 = m$ then how fast will q_2 be moving when it is at a distance $\frac{c}{4}$ from the origin (ignore gravity)

Target: v

method: $v_{fy}^2 = v_{0y}^2 + 2|\vec{a}_y| * |\Delta \vec{y}|$ ↗ a isn't constant

method: $W_{net} = k_f - k_0$ (problem is about two speed at two points)

$$\underbrace{W_{Fe}}_{-\Delta u_e} = k_f - k_0 = \frac{1}{2}mv^2$$

$$-\Delta u_e = k_f - k_0$$

$$-q_0 \underbrace{\Delta V_{\text{source(s)}}}_{\text{electric pot.}} = k_f - k_0$$

$$\underbrace{V_i}_{\substack{\text{is relative to} \\ V=0 \text{ at } r=\infty}} = V_{1i} + V_{2i}$$

$$= \frac{kq_1}{r_{1i}} + \frac{kq_2}{r_{2i}}$$

$$= \frac{k2q}{\sqrt{a^2+c^2}} + \frac{k2q}{\sqrt{a^2+c^2}}$$

$$= \frac{4kq}{\sqrt{a^2+c^2}}$$

$$= V_{1f} + V_{2f}$$

$$= \frac{kq_1}{r_{1f}} + \frac{kq_2}{r_{2f}}$$

$$= \frac{2kq}{\sqrt{a^2+(\frac{c}{4})^2}} + \frac{2kq}{\sqrt{a^2+(\frac{c}{4})^2}}$$

$$= \frac{4kq}{\sqrt{a^2+(\frac{c}{4})^2}}$$

$$-(-q)4kq \left[\frac{1}{\sqrt{a^2 + (\frac{c}{4})^2}} - \frac{1}{\sqrt{a^2 + c^2}} \right] = \frac{1}{2}(3m)v^2$$

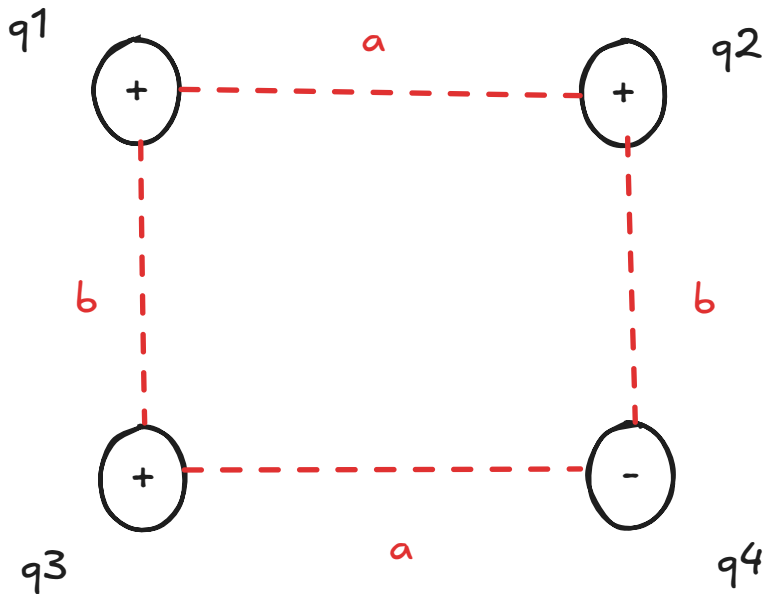
$$\Rightarrow v^2 = \frac{8kq^2}{3m} \left[\frac{1}{\sqrt{a^2 + (\frac{c}{4})^2}} - \frac{1}{\sqrt{a^2 + c^2}} \right]$$

$$\Rightarrow v = \sqrt{\frac{8kq^2}{3m} \left[\frac{1}{\sqrt{a^2 + (\frac{c}{4})^2}} - \frac{1}{\sqrt{a^2 + c^2}} \right]}$$

#date/2025/10/17

the concept of electric energy stored in a configuration of charges

you place 4 point charges $q_1, q_2, q_3,$ and, q_4 as shown, calculate the energy stored in this configuration



where: $q_1 = q, q_2 = 2q, q_3 = 3q, q_4 = -4q$

Energy stored in a system = the work done to assemble "the system"

$$W_{\text{by us}} + W_{\text{electric}} = \cancel{\Delta K}^0$$

$$W_{\text{by us}} + [-\Delta u_{\text{electric of system}}] = 0$$

$$\downarrow$$

$$\epsilon_{\text{stored by us}} = \Delta u_{\text{electric of system}}$$

$$\begin{aligned} \epsilon_{\text{stored}} &= 0 + q_2 V_{12} + q_3 [V_{13} + V_{23}] + q_4 [V_{14} + V_{24} + V_{34}] \\ &= q_2 \frac{kq_1}{r_{12}} + q_3 \left(\frac{kq_1}{r_{13}} + \frac{kq_2}{r_{23}} \right) + q_4 \left(\frac{kq_1}{r_{14}} + \frac{kq_2}{r_{24}} + \frac{kq_3}{r_{34}} \right) \\ &= 2kq^2 + 3q \left[\frac{kq}{b} + \frac{k(2q)}{\sqrt{a^2 + b^2}} \right] + (-4q) \left(\frac{kq}{\sqrt{a^2 + b^2}} + \frac{k(2q)}{b} + \frac{k(3q)}{a} \right) \end{aligned}$$

note from dhar: theres a very painful homework problem where the point charges are arranged in an equilateral triangle, which is good prep for midterm

Now suppose. $q_1 = q_2 = q_3 = q_4 = -7Q_0$ and $a = b$, also $m_1 = m_2 = m_3 = m_4 = M$

if you release all of them simultaneously, how fast would they be moving when they are very far away

hint from problem statement: 2 speeds at 2 points (initially 0, want to find V_f)

target: v_∞

method:

$$W_{net} = k_{\infty} - k_i$$

$$-[\cancel{u_{\infty}}^0 + u_i] = k_{\infty} - \cancel{k_i}^0$$

$$\underbrace{u_i}_{u_{stored}} = k_{\infty}$$

$$u_{\text{stored}} = \frac{49kQ_0^2}{a} + 49kQ_0^2 \left[\frac{1}{b} + \frac{1}{\sqrt{a^2 + b^2}} \right] + 49kQ_0^2 \left[\frac{1}{\sqrt{a^2 + b^2}} + \frac{1}{b} + \frac{1}{a} \right]$$

$$= 49kQ_0^2 \left[\frac{2}{a} + \frac{2}{b} + \frac{2}{\sqrt{a^2 + b^2}} \right]$$

$$= \frac{49kQ_0^2}{a} [4 + \sqrt{2}] = u_{\infty} = k_i$$

$$\frac{49kQ_0^2}{a} [4 + \sqrt{2}] = \frac{1}{2} (4M) v_{\infty}^2$$

$$v_{\infty}^2 = \frac{98qkQ_0^2}{4Ma} [4 + \sqrt{2}]$$

$$v_{\infty} = \sqrt{\frac{98qkQ_0^2}{4Ma} [4 + \sqrt{2}]}$$